

SANGAR, Russian FederationWMO: **246520**Lat: **63.959N**Lon: **127.421E**Elev: **314**StdP: **14.53**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-54.6	-51.7	-58.0	0.2	-54.5	-55.1	0.2	-51.6	22.8	-16.4	19.1	-19.9	1.4	300	0.761

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	82.5	65.4	79.1	64.0	75.7	62.4	68.1	77.9	66.0	75.3	64.1	73.1	6.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	91.8	72.2	62.4	85.2	71.3	60.3	79.1	69.1	32.7	77.6	31.0	75.4	29.6	73.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	18.9	16.4	DB	-56.7	87.9	3.5	3.2	-59.2	90.2	-61.3	92.0	-63.3	93.8	-65.8	96.2	(4)
(5)			WB	-56.7	71.9	3.5	2.4	-59.2	73.7	-61.3	75.1	-63.3	76.4	-65.8	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	-35.1	-27.3	-4.1	21.4	41.9	59.2	65.1	58.6	41.8	17.6	-16.6	-34.1
	DBStd	37.83	10.45	10.44	13.41	10.95	8.83	7.86	6.82	6.67	7.55	11.15	14.23	11.21
	HDD50	13518	2638	2164	1676	859	282	16	1	9	262	1004	1998	2609
	HDD65	18051	3103	2584	2141	1309	718	204	86	219	696	1469	2448	3074
	CDD50	1079	0	0	0	0	29	291	468	275	16	0	0	0
	CDD65	138	0	0	0	0	0	30	88	20	0	0	0	0
	CDH74	1083	0	0	0	0	0	271	694	118	0	0	0	0
	CDH80	219	0	0	0	0	0	44	161	14	0	0	0	0
Wind	WSAvg	6.9	4.9	5.2	6.6	8.0	8.4	7.9	7.3	7.1	7.7	7.8	6.5	4.9
Precipitation	PrecAvg	13.3	0.4	0.4	0.5	0.6	1.1	1.6	2.0	2.0	1.8	1.7	0.8	0.5
	PrecMax	18.6	1.0	1.3	1.5	1.7	2.6	5.2	5.0	4.6	4.7	3.4	2.0	1.2
	PrecMin	7.8	0.1	0.0	0.0	0.1	0.0	0.2	0.4	0.2	0.0	0.3	0.2	0.1
	PrecStd	2.6	0.3	0.3	0.4	0.3	0.6	1.0	1.2	1.0	1.1	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.2	4.8	32.9	50.1	70.1	84.6	87.9	81.8	66.9	40.5	19.5	-3.7
		MCWB	-6.0	3.7	28.2	40.4	51.4	64.6	67.5	66.7	55.2	35.9	17.6	-4.4
	2%	DB	-10.7	0.2	26.6	45.2	65.3	80.6	84.3	77.1	60.9	35.9	13.6	-8.8
		MCWB	-11.3	-0.7	23.3	36.5	49.7	62.7	66.3	63.8	51.9	32.8	12.5	-9.4
	5%	DB	-15.3	-5.6	21.6	41.0	60.8	76.9	81.0	73.9	57.1	33.5	9.3	-12.7
		MCWB	-15.7	-6.4	18.7	34.0	47.6	60.6	65.6	61.9	49.8	31.2	8.4	-13.2
	10%	DB	-19.7	-10.9	16.3	37.3	56.5	73.4	77.8	70.5	53.5	31.1	5.0	-17.3
		MCWB	-20.1	-11.4	14.0	31.3	45.2	59.0	64.5	60.3	47.4	29.3	4.1	-17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.8	3.9	28.3	40.4	53.2	68.3	71.7	68.6	56.6	37.0	18.0	-4.3
		MCDB	-5.2	4.6	32.6	49.0	64.8	80.7	82.7	78.3	63.9	39.8	19.6	-3.7
	2%	WB	-11.2	-0.8	23.3	37.0	50.9	64.7	69.2	65.6	53.7	33.3	12.7	-9.3
		MCDB	-10.7	0.1	26.7	43.8	62.9	76.1	79.8	74.1	58.7	35.0	13.6	-8.8
	5%	WB	-15.7	-6.4	18.7	34.4	48.5	62.6	67.4	63.5	50.9	31.4	8.4	-13.1
		MCDB	-15.3	-5.7	21.5	39.9	58.7	73.4	76.8	71.0	56.1	33.3	9.3	-12.7
	10%	WB	-20.0	-11.4	14.1	32.0	46.2	60.7	65.8	61.4	48.3	29.2	4.2	-17.6
		MCDB	-19.7	-10.9	16.4	36.9	55.4	71.0	74.7	68.7	52.7	31.1	5.0	-17.3
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.2	17.5	18.8	15.7	17.3	16.7	15.7	13.0	9.0	10.2	9.0
		MCDBR	14.4	14.1	19.0	18.4	20.5	23.3	21.6	20.4	19.4	9.4	11.9	12.4
		MCWBR	14.2	13.4	16.2	12.3	11.0	11.2	10.0	10.4	11.7	7.2	11.3	12.1
	5% WB	MCDBR	14.4	13.9	18.6	16.7	18.7	19.8	18.7	17.4	16.9	8.8	11.8	12.4
		MCWBR	14.1	13.3	15.9	11.5	10.5	10.7	9.9	9.8	10.9	7.1	11.2	12.1
Clear-Sky Solar Irradiance	taub		0.193	0.248	0.293	0.345	0.399	0.414	0.492	0.472	0.344	0.270	0.193	0.180
	taud		1.910	2.077	2.077	2.072	2.191	2.265	2.026	2.027	2.381	2.266	1.921	1.884
	Ebn at Noon		121	212	248	259	256	254	226	215	234	204	115	52
	Edh at Noon		9	20	31	40	39	37	47	43	24	16	9	4
All-Sky Solar Radiation	RadAvg		67	350	925	1582	1700	1886	1712	1269	757	300	110	25
	RadStd		3	16	40	67	150	125	84	107	106	37	9	1